

An AI-Based System for Cyberbullying Detection: A Comparative Study of Machine Learning and Deep Learning Approaches

Literature Survey

Introduction

Social media has become an inseparable part of daily life, connecting people across the globe. However, this connectivity has also provided a space for negative behaviors, including online harassment, abusive comments, and threatening messages. This phenomenon, cyberbullying, poses a serious threat to the mental health and confidence of individuals. In response, a significant body of research has focused on creating systems that can automatically recognize such harmful content using Machine Learning (ML), Deep Learning (DL), and Natural Language Processing (NLP).

Despite this progress, significant limitations remain. A review of the current landscape reveals two critical gaps:

1. **A Language and Context Gap:** Most existing models are built and trained only for English text. They often fail when processing the complex, real-world language of social media, which includes slang, short forms, emojis, and code-mixed languages like Hinglish.
2. **A Practicality Gap:** The majority of research focuses on model performance in a lab setting. There are very few tools available that can work directly on a user's browser in real time to offer instant protection.

This literature review surveys the key methodologies in cyberbullying detection, from traditional machine learning to state-of-the-art transformers. The primary objective is to critically analyze their strengths and weaknesses to build a foundation for a new system that directly addresses the identified gaps: a practical, real-time browser extension capable of handling both English and Hinglish content.

Review of Methodologies

Traditional Machine Learning Approaches (TF-IDF, BoW)

Early and baseline approaches for cyberbullying detection have relied heavily on traditional machine learning classifiers paired with feature extraction techniques like TF-IDF and bag-of-words (BoW).

- **Rahman et al. (2021) [1]** used a combined social-media dataset from sources like Kaggle and Twitter. They extracted TF-IDF features and tested SVM, logistic regression, random forest (RF), and Naïve Bayes. On the baseline imbalanced data, SVM and logistic regression achieved ~88% accuracy. However, after applying SMOTE oversampling, a Random Forest achieved the best result with 89% accuracy and 89% F1-score.
- **Dhumale et al. (2025) [2]** proposed a framework using both Word2Vec embeddings and TF-IDF. On their self-compiled English corpus, an SVM classifier using skip-gram Word2Vec vectors achieved the highest accuracy at ~95%, significantly outperforming models using CBOW (93%) and other classifiers.
- **Dewani et al. (2023) [3]** focused on the low-resource challenge of Roman Urdu from Twitter. They compiled a domain-specific lexicon of profanity and slang. Using TF-IDF features (unigrams+ngrams), an SVM achieved the highest accuracy of ~83%, demonstrating that traditional features+SVM can yield strong results for under-resourced languages.

- **General Finding:** Surveys note that supervised ML, particularly SVM, is "particularly effective" in classic cyberbullying tasks [4]. When paired with TF-IDF or BoW, SVMs frequently provide strong baseline performance, with many works reporting 80-90% accuracy.

Deep Learning Approaches (CNN, LSTM, Bi-LSTM)

With the rise of deep learning for NLP, researchers moved to architectures capable of learning features automatically from text sequences.

- **Sultan et al. (2023) [5]** designed a hybrid CNN-LSTM model. Trained on a large multi-platform corpus (Twitter, YouTube), their model achieved 97.5% accuracy. This result far surpassed their ML baselines (e.g., Random Forest at ~58% acc), highlighting the power of DL models to capture complex features.
- **Handayani & Abas (2024) [6]** compared a CNN-LSTM stack against a vanilla LSTM on datasets of increasing size. They found that the CNN-LSTM hybrid progressively outperformed the standalone LSTM as the dataset grew, suggesting that convolutional feature-extraction adds significant value given sufficient data.
- **Aldhyani et al. (2022) [7]** studied both binary and multi-class cyberbullying on Twitter. In their experiments, a pure Bi-directional LSTM (BiLSTM) model outperformed their proposed CNN-BiLSTM hybrid, reaching 99% accuracy on a 6-class task.
- **General Finding:** Deep learning methods consistently outperform traditional baselines on large datasets [4]. Architectures like CNN, LSTM, BiLSTM, and GRU are widely used, often paired with pre-trained word embeddings like Word2Vec or GloVe, which has been shown to boost F1-scores by as much as 18%.

Transformer-based Models (BERT, RoBERTa)

The current state-of-the-art in NLP is dominated by large pre-trained transformer models, which leverage contextual language understanding.

- **Ogunleye & Dharmaraj (2023) [8]** fine-tuned BERT-family models on a new English dataset from Formspring and Twitter. They reported that **RoBERTa** outperformed all other models, confirming the power of large pre-trained models in capturing nuanced abusive language.
- **Kumar et al. (2025) [9]** tackled code-mixed Hinglish (Hindi-English) cyberbullying. They used **MURIL** (Multilingual Representations for Indian Languages) and evaluated it across six heterogeneous datasets. The MURIL-based model consistently outperformed other multilingual models, achieving 86.97% accuracy on the Bohra et al. Hinglish dataset compared to 82.45% for RoBERTa. This strongly demonstrates the benefit of using models pre-trained specifically on code-mixed Indian languages.
- **Maity et al. (2024) [10]** introduced BullyExplain, a new code-mixed dataset with rationale annotations. They proposed a *transformer-free* model, mExCB (a CNN+GRU multi-task model with attention), which was effective on code-mixed data and could also provide explanations.
- **General Finding:** BERT-based models are now state-of-the-art. Even distilled transformers (DistilBERT) outperform classical ML [4]. In multilingual contexts, models like **XLM-RoBERTa** often give the highest scores [11].

Advanced and Multimodal Methods

Recent work has focused on the specific challenges of real-world social media data, such as slang, code-mixing, and emojis.

- **Patel et al. (2025) [12]** developed a system that analyzes both text and emoji sentiment. They used an ensemble of ML and DL models trained on TF-IDF features that included emoji sentiment scores. A key contribution was a severity score (1-5) to automatically trigger user blocking, creating a proactive filter.
- **Rahman-Laskar et al. (2024) [13]** created MC-Hinglish1.0, a new multi-class dataset for code-mixed Hinglish with seven distinct bullying categories (e.g., age, gender, religion). Their work highlights the challenge of fine-grained classification in code-mixed languages.
- **Dewani et al. (2023) [3]**, in their work on Roman Urdu, emphasized the importance of advanced preprocessing. They compiled dictionaries of Urdu slang and normalized spellings, a pipeline that improved ML performance by effectively handling informal language.

Practical Systems and Tools

A final trend is the translation of these models into user-facing tools.

- **Zhang & Park (2023) [14]** developed a real-time Chrome extension for child safety. It uses a fine-tuned **BERT** to flag abusive text and a **YOLOv8 CNN** to detect harmful images, automatically blocking content. They note that this requires GPU acceleration for low-latency filtering.
- This illustrates the clear trend of packaging top-performing detection models (like BERT and CNNs) into browser add-ons and moderation tools to filter content in real time.

Conclusion: Summary and Identified Research Gap

This review confirms a clear evolution in cyberbullying detection. The field has progressed from traditional ML models like SVM, which serve as strong baselines, to more powerful deep learning models like CNN-LSTM hybrids that can capture complex patterns. The current state-of-the-art is held by large transformer models, with RoBERTa dominating English tasks and specialized multilingual models like MURIL showing superior performance on code-mixed Hinglish text [9].

Despite this progress, this review identifies two critical gaps, which form the basis of this project:

1. **The Language Gap:** The vast majority of models are designed for and tested on clean, English-only datasets. As shown by Kumar et al. (2025) [9] and Dewani et al. (2023) [3], these general models perform poorly on the messy, real-world data found on social media, which is rich in slang, emojis, and code-mixed languages like Hinglish. While studies on MURIL and Roman Urdu are emerging, dedicated support for Hinglish and slang in a practical tool is still lacking.
2. **The Practicality Gap:** Most research ends with a performance metric in a paper. While a few practical tools exist, such as the Chrome extension by Zhang & Park (2023) [14], there is a distinct lack of accessible, real-time tools that are lightweight and effective for the average user

Project Justification

This project aims to directly address these gaps by bridging the divide between theoretical performance and practical utility. The primary objective is to develop a tool that is not only accurate but also practical.

This will be achieved by pursuing two main goals. First, the project will conduct a comprehensive **comparative study** of various algorithms (SVM, RF, LSTM, CNN, and BERT-based models) to determine which algorithm provides the best balance of accuracy and speed for this task. Second, it will use this finding to build a **practical browser extension**. This tool will be explicitly designed to overcome the limitations identified in the literature by incorporating:

1. **Hinglish and Slang Support:** Moving beyond English-only detection.
2. **Severity Scoring:** Classifying content as mild, aggressive, or severe to provide more nuanced filtering.

By creating a user-friendly tool that functions in real time, this project will provide both a research contribution (the comparative analysis) and a tangible solution that contributes to making the internet a safer and more respectful place.

References

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